

Does Autoplay Drive Excessive Screen Time? Evidence from an Online experiment

1-hour presentation

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Motivation



- Autoplay is a ubiquitous design feature delivering video content without explicit consent
- It is a product of the extensive A/B testing done by companies to increase engagement

Motivation

- Companies maximize content consumption via (1) proprietary recommendation algorithms, (2) interface design
- Despite subsequent legislative action in US (SMART Act of 2019) and in EU (Digital Fairness Act), causal effects of autoplay remain unclear
- In order to inform policy, welfare implications of interface design needs deeper study
- This study is an (online) laboratory experiment isolating autoplay's effect on content consumption

Research Question

- Does autoplay, in isolation from algorithmic content curation, increase video consumption and override users' stated preferences?

Related Literature

1. Digital addiction and welfare effects of social media
 - Allcott et al. (2020, 2022); Braghieri et al. (2022); Bursztyn et al. (2023)
2. Intertemporal choice and self-control
 - Bonein and Denant-Boèmont (2015); Ek and Samahita (2023); Ericson and Laibson (2019); Houser et al. (2018)
3. Interface design and user agency
 - Lukoff et al. (2021); Lyngs et al. (2020); Mathur et al. (2021); Purohit et al. (2023); Schaffner et al. (2023)
4. Field evidence on autoplay
 - Hiniker et al. (2018); Schaffner et al. (2025)

Contribution

1. Main: First study to causally identify autoplay's effect on content consumption
2. Methodological: Intertemporal choice with real-time labor-leisure tradeoff
3. Policy: Lessons for regulation

Theoretical Framework

- Temptation in digital context, as in Kleinberg et al. (2022)
- Preregistered Hypotheses (<https://osf.io/mfkbc/overview>)
 1. Autoplay increases video consumption
 2. Autoplay causes more preference reversals
 3. Participants demand commitment against autoplay

Preview of Results

- Autoplay shows no causal effect on video consumption when content is held constant
- Participants exhibit positive Willingness To Pay (WTP) for autoplay (6.72 pence/hour), viewing it as a convenience feature rather than a self-control problem

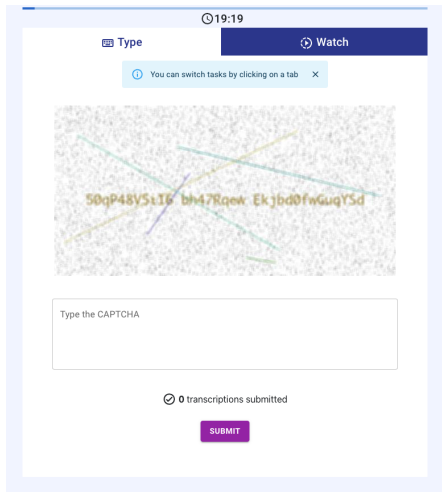
Experimental Design Overview

Timeline	
Block 1	Practice → Time Choice → Main Session
Block 2	Practice → Multiple Price List → Time Choice → Main Session

- Two separate data collections with separate samples (Blocks 1 & 2)
- Objective: Time allocation across a labor and a leisure task
- 2-minute Practice Session to familiarize with tasks
- Time Choice: a hypothetical time allocation decision
- Main session: 20 minutes of time to be spent between tasks
- Block 2: Multiple Price List (MPL) to elicit WTP for autoplay

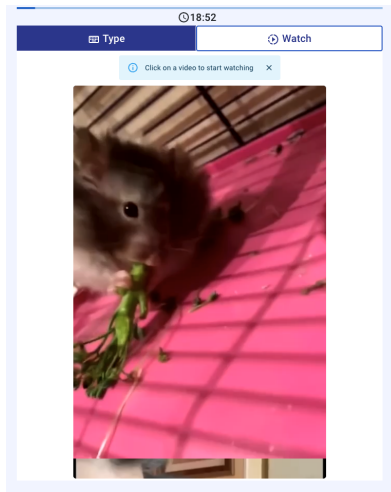
Transcription Task

- Retype randomly generated CAPTCHAs (35 characters, blurred and distorted)
- Payment at 0.15 pence/second
- Quality requirements at 70% accuracy, minimum 1 submission/minute
- Average time 41 seconds per CAPTCHA



Watching Task

- Watch funny animal videos curated from YouTube and TikTok
- Payment at 0.10 pence/second
- 80 videos total, short duration (mean 12.25 sec.)
- Videos rated positively (median 7/10)



Treatment Variation

	Autoplay	Control
Playback	Videos play continuously	Participants click to play each video
Transition	Automatic scrolling to next video	Next video visible but requires click

- Identical videos and playlist order
- Same tab structure, progress bar, and timer
- Not possible to skip videos (controls disabled)
- Randomization occurs before Practice session

Multiple Price List

- Only in Block 2
- 9 binary decisions
- Allows probabilistically selecting into treatment or control
- Bonus earnings range between 0.5 and 0.05 pounds
- 1 out of 9 choices implemented at random

Autoplay on or off?

In the practice session videos played automatically with **Autoplay**. There is another version **where you need to click on videos to play them**.

You need to make 9 decisions about Autoplay. One of your decisions will be randomly chosen and implemented. **You will receive a bonus payment and watch videos with the chosen Autoplay setting.**

For example, if you choose Autoplay for the first decision and it is implemented, you will have Autoplay and receive an additional £0.5 bonus payment.

 AUTOPLAY +£0.5	 NO AUTOPLAY +£0
 AUTOPLAY +£0.25	 NO AUTOPLAY +£0
 AUTOPLAY +£0.1	 NO AUTOPLAY +£0
 AUTOPLAY +£0.05	 NO AUTOPLAY +£0
 AUTOPLAY +£0	 NO AUTOPLAY +£0
 AUTOPLAY +£0	 NO AUTOPLAY +£0.05
 AUTOPLAY +£0	 NO AUTOPLAY +£0.1
 AUTOPLAY +£0	 NO AUTOPLAY +£0.25
 AUTOPLAY +£0	 NO AUTOPLAY +£0.5

 Please pick your favorite option for each row.

CONTINUE

Time Choice

- Two-day online experiment
- 24-hour cooling off period to measure planned vs. actual behavior
- Real-time feedback on expected earnings with slider tool (20-second increments)
- Corner solutions allowed (0% or 100% on either task) with max. earnings difference at 0.6 pounds
- 5% of choices randomly made binding for Day 2

Time Choice

Now decide how long you would like to spend **tomorrow** on **Typing** and on **Watching Videos**.

Tomorrow you will have **20 minutes** for both tasks.

1 out of every 20 participant will be **bound** by the choice they made today.

Please **click on the slider** and indicate your **Time Choice**:



No Typing Equal split Only Typing

① You choose to spend **10 minutes 0 seconds** to **Type**.
You choose to spend **10 minutes 0 seconds** to **Watch Videos**.
You would earn **£4.25**.

Tomorrow you will learn whether your **Time Choice** is binding.

CONTINUE

Experimental Design Recap

- Key tradeoff: higher earnings from Transcription vs. lower effort from Watching
- Elicit informed time allocation preferences before actualization to measure preference reversals in any direction
- Treatment only changes the autoplay setting in Watching task

	Day 1	Day 2
Block 1	Practice → Time Choice	Main Session
Block 2	Practice → MPL → Time Choice	Main Session

Procedures

- Prolific participant pool (online platform for academic research)
- Adult British residents, fluent in English
- Sampling based on preregistered power calculations (<https://osf.io/mfkbc/overview>)
- 276/301 participants completed Block 1 (92% response rate)
- 57/60 participants completed Block 2 (95% response rate)
- Fixed fee: £2.75 upon completion
- Average total earnings: £4.08
- Average total time spent: 30 minutes

Internal Controls

- 2 attention checks
- Desktop only, Chromium browsers on both days
- Minimum internet speed >30 Mbps on both days
- Real-time mouse monitoring using JavaScript
- Post-experiment validation (self-reported honesty and network quality)
- Exclusion criteria: Connection issues (6%), honesty (9%), mouse monitoring (15%)



Figure: Experience Day 1 as a visitor!

Balance table Block 1

	Autoplay N = 79	Control N = 105	<i>p</i> [A = C]
Gender			
Female	36 (45.57%)	53 (50.48%)	0.510
Age			
Mean (SD)	42.63 (11.70)	42.03 (11.24)	0.725
Employment			
Employed (full time)	45 (56.96%)	57 (54.29%)	0.718
Employed (part time/self)	23 (29.11%)	25 (23.81%)	0.417
Not employed	11 (13.92%)	23 (21.90%)	0.167
Income			
Low (0-15)	24 (30.38%)	32 (30.48%)	0.989
Middle (15-50)	42 (53.16%)	61 (58.10%)	0.505
High (50+)	13 (16.46%)	12 (11.43%)	0.325
Marital Status			
Married/Partnership	43 (54.43%)	52 (49.52%)	0.510
Single	31 (39.24%)	46 (43.81%)	0.534
Previously married	5 (6.33%)	7 (6.67%)	0.927

Balance table Block 2

	Main Sample N = 184	MPL N = 52	<i>p</i> [M = MPL]
Gender			
Female	89 (48.37%)	26 (50.00%)	0.835
Age			
Mean (SD)	42.29 (11.41)	41.90 (14.14)	0.858
Employment			
Employed (full time)	102 (55.43%)	32 (61.54%)	0.433
Employed (part time/self)	48 (26.09%)	10 (19.23%)	0.311
Not employed	34 (18.48%)	10 (19.23%)	0.902
Income			
Low (0-15)	56 (30.43%)	15 (28.85%)	0.825
Middle (15-50)	103 (55.98%)	30 (57.69%)	0.826
High (50+)	25 (13.59%)	7 (13.46%)	0.981
Marital Status			
Married/Partnership	95 (51.63%)	28 (53.85%)	0.778
Single	77 (41.85%)	19 (36.54%)	0.491
Previously married	12 (6.52%)	5 (9.62%)	0.446

Sample statistics

	Autoplay N = 79	Control N = 105	p [A = C]
<i>Time Choice (Planned)</i>			
Mean (SD)	0.48 (0.31)	0.44 (0.33)	0.410
Median [Min, Max]	0.50 [0.00, 1.00]	0.50 [0.00, 1.00]	
<i>Nb. of sessions</i>			
Mean (SD)	5.76 (4.98)	4.89 (5.09)	0.246
Median [Min, Max]	4.00 [1.00, 18.00]	3.00 [1.00, 28.00]	
<i>Session length</i>			
Mean (SD)	456.11 (383.22)	546.35 (411.53)	0.131
Median [Min, Max]	314.75 [68.17, 1212.00]	418.67 [44.93, 1314.00]	
<i>Content rating</i>			
Mean (SD)	6.96 (2.18)	6.62 (2.45)	0.327
Median [Min, Max]	7.00 [2.00, 10.00]	7.00 [0.00, 10.00]	

H1: Autoplay and video consumption

Hypothesis 1: Autoplay increases video consumption.

$$Y_i = \beta_0 + \beta_1 \text{Autoplay}_i + \beta_2 \text{TimeChoice}_i + \beta_3 (\text{Autoplay}_i \times \text{TimeChoice}_i) + \epsilon_i \quad (1)$$

- where Y_i represents the standardized number of videos watched
- Autoplay_i is a binary indicator for the condition
- TimeChoice_i is the standardized proportion of time participants planned to watch
- Interaction term examines the effect of autoplay based on initial time allocation plans
- H1 posits $\beta_1 > 0$ for videos watched

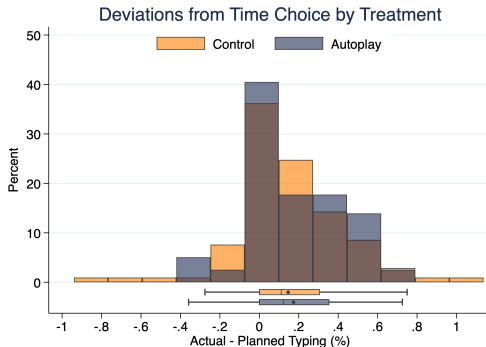
H1: Autoplay and video consumption

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	-0.053 (0.149)	0.029 (0.112)	0.031 (0.112)
Time Choice (z-score)		0.664*** (0.056)	0.619*** (0.079)
Autoplay $\times$ Time Choice			0.114 (0.105)
Constant	0.023 (0.098)	-0.012 (0.076)	-0.010 (0.076)
Obs.	184	184	184
R^2	0.001	0.440	0.443
F-statistic	0.13	73.06	61.53

Note: Robust standard errors in parentheses. All variables standardized. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

H2: Autoplay and preference reversals

- **Hypothesis 2:** Autoplay causes more preference reversals.
- A two-sample t -test reveals no statistically significant difference between means ($t = -0.694$, $p = 0.488$)
- Kolmogorov-Smirnov test similarly finds no significant differences in distributions ($D = 0.085$, $p = 0.903$)



H3: Commitment against autoplay

Hypothesis 3: There is positive demand for a commitment device against autoplay.

$$Y_{ij} = \beta_0 + \beta_1 \text{Bonus}_{ij} + \epsilon_{ij} \quad (2)$$

- where Y_{ij} is the binary choice of taking autoplay given the bonus j from the MPL for individual i
- WTP for autoplay as the sample-averaged effect of bonus $-\beta_0/\beta_1$
- H3 posits $-\beta_0/\beta_1 > 0$ for preference against autoplay

H3: Commitment against autoplay

- WTP for autoplay is the negative ratio of the constant to the bonus coefficient:
 $-0.6263/27.9729 = -0.0224$
pounds, or 2.24 pence for the 20-minute session
- Participants value autoplay at 6.72 pence/hour and would be willing to forgo this amount to maintain access to the feature

	Autoplay (1)
Bonus (£)	27.973*** (6.277)
Constant	0.626** (0.199)
Obs.	468
Clusters	52
Pseudo R^2	0.658
Wald $\chi^2(1)$	19.86

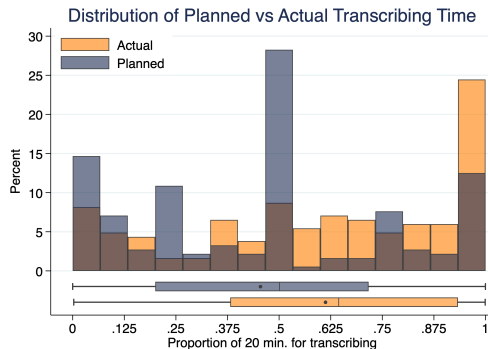
Note: Logistic regression with standard errors clustered at the participant level in parentheses. Dependent variable is binary choice of selecting autoplay (1) or turning it off (0). Bonus amounts range from £−0.5 to £0.5. The pseudo R^2 indicates good fit. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Main Findings Recap

- Autoplay did not increase the nb. of video content consumed nor caused more preference reversals (H1 & H2)
- Participants actually deviated from their time allocation plans toward more Transcription!
- Participants are willing to pay for the Autoplay feature (H3)

Mechanisms

- Time inconsistent behavior toward more earnings (max. 60 pence) on Day 2
- Why exert more effort than initially planned for?

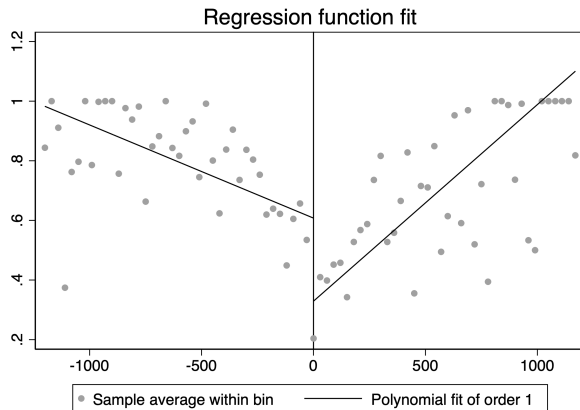




- Transcription task was perceived negatively (29%)
- Watching videos as taking breaks from transcription (23%)
- Unintentional “work environment” due to experimenter demand effects

Achieving Time Allocation Plans

- Regression Discontinuity with 30-second bins around planned time allocation
- Participants aim to achieve their plans...
- ...but eventually they get back to “work”



Conclusion

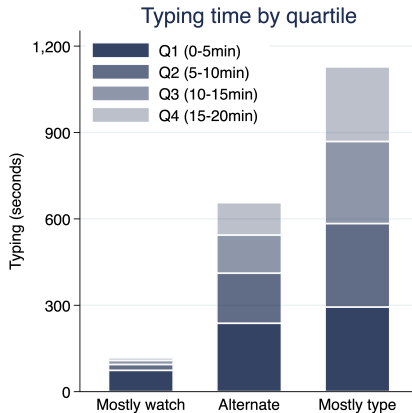
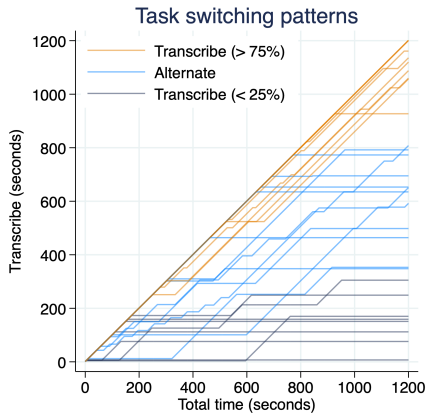
- First study to causally identify autoplay's effect on content consumption
- Methodological: unexpected within the intertemporal choice framework (implementation)
- Causes: study design (earnings difference, focus on transcription requirements), quirks of the Prolific participant pool (research for a living), postponing leisure after the experiment (spoonfeeding leisure)
- Policy: more suitable for field work, albeit more intrusive and technically complex

Why pay for Autoplay?

	Control	Autoplay	Difference
Total seconds lost	46.52 (9.27)	1.08 (0.65)	45.45*** (9.29)
Observations	105	79	184
<i>t</i> -statistic			4.89

- Efficiency gains
- Easing consumption

Robustness: Participant Behavior



Robustness: Full Sample

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	0.051 (0.135)	0.131 (0.109)	0.132 (0.109)
Time Choice (z-score)		-0.590*** (0.060)	-0.555*** (0.084)
Autoplay $\times$ Time Choice			-0.077 (0.118)
Constant	-0.023 (0.090)	-0.061 (0.075)	-0.059 (0.075)
Obs.	224	224	224
R^2	0.001	0.347	0.348
F-statistic	0.14	48.91	34.21

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Full RD table

	Time Window		
	30-seconds	60-seconds	90-seconds
Proportion transcribing	-0.274*** (0.101)	-0.255*** (0.087)	-0.075 (0.108)
Obs. (left)	66,786	67,985	67,119
Obs. (right)	80,270	89,006	92,999
Clusters (left)	157	154	155
Clusters (right)	154	157	161
Bandwidth	339.9	368.3	422.3

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